

Ground Zero: The Chernobyl Response Group

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High School Specialized Agency

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Disaster Background:

On 26 April 1986, the fourth reactor unit at the Chernobyl Nuclear Power Plant, located near the city of Pripyat in the Ukrainian Soviet Socialist Republic, suffered a catastrophic explosion and meltdown during a late-night safety test. The disaster released huge quantities of radioactive material across Ukraine, Belarus, and parts of the eastern Soviet Union and Europe. Design flaws in technology combined with operational errors and a deficient safety culture played a key role in causing the accident. The Soviet government's initial secrecy, delay in public communication, and inadequate evacuation planning further amplified the humanitarian, environmental, and political consequences.

You are tasked with responding to the immediate, high-stakes operational issues presented by the Chernobyl disaster directly following its occurrence in 1986. The survival of the Soviet Union and the European continent rests on your immediate decisions.

Topic A: The 48-Hour Threshold: Containment & The Thermal Threat

Background:

On April 26, 1986, an explosion occurred at Reactor No. 4 of the Chernobyl Nuclear Power Plant in Pripyat. The accident was the result of a flawed RBMK-1000 reactor design, specifically its positive void coefficient and graphite-tipped control rods, combined with human error during a late-night safety test. The initial steam explosion blew the 1,000-ton biological shield off the reactor, completely exposing the nuclear core to the open air. Because the reactor utilized graphite as a neutron moderator, the introduction of oxygen ignited the graphite, creating an unprecedented open-air nuclear fire that acts as a massive chimney, pumping highly radioactive isotopes into the upper atmosphere.

Beyond the immediate atmospheric contamination, a second, infinitely more destructive threat exists. Beneath the burning reactor core lie the "bubbler pools"—massive suppression tanks filled with millions of liters of water designed to condense steam in the event of a broken coolant pipe. With the core now exposed and melting, a superheated, radioactive lava is forming. If this 2,500-degree Celsius lava burns through the remaining concrete floor and drops into the flooded bubbler pools, the rapid expansion of water into steam will trigger a second, massive thermal explosion. Experts estimate this secondary steam blast could yield a force of up to 4 megatons. An explosion of this magnitude would vaporize the remaining three intact reactors at the Chernobyl facility, instantly destroying the city of Pripyat and dispersing enough lethal radioactive material to render a massive portion of Eastern Europe entirely uninhabitable for thousands of years.

Current Situation:

The committee convenes in the immediate hours following the explosion. The environment is actively hazardous; the destroyed reactor is spewing radioactive material, while plant workers and local residents remain largely unaware of the extreme danger. The graphite fire is burning out of control, and standard firefighting techniques utilizing water have proven not only ineffective but actively dangerous, as water vaporizes instantly and exacerbates the release of radioactive steam. First responders, including local firefighters and plant operators, are absorbing fatal doses of radiation in a desperate, blind attempt to control the facility.

Simultaneously, the clock is ticking on the thermal threat. The bubbler pools remain full, and the exact temperature and trajectory of the melting core are difficult to ascertain due to the destruction of the plant's diagnostic sensors. The immediate challenge is twofold: the graphite fire must be smothered from above to halt the atmospheric spread, and the water must be drained from below to prevent the secondary explosion.

Mass evacuation protocols for the city of Pripyat (population 50,000) have not yet been initiated, leaving civilians exposed to catastrophic levels of Iodine-131 and Cesium-137. The committee must grapple with an unprecedented nuclear catastrophe where every passing hour limits the available options and increases the mortality rate.

Past Actions:

Following the explosion, the immediate actions taken on the ground relied on standard Soviet civil defense and firefighting protocols, which were wholly inadequate for an exposed nuclear core. Plant management initially refused to believe the core had exploded, reporting to Moscow that the reactor was intact and the issue was merely a roof fire. This denial led to the deployment of firefighters without radiological protective gear, resulting in mass casualties from Acute Radiation Syndrome (ARS).

Historically, the Soviet Union has relied on strict, centralized chains of command during industrial accidents, limiting the flow of information to prevent panic. However, there is no precedent or established protocol for an open-air reactor fire. Previous nuclear accidents within the USSR (such as the Kyshtym disaster in 1957) were kept entirely secret, but the scale of the Chernobyl explosion has rendered standard concealment tactics obsolete. The state lacks the specialized machinery to clear lethal debris or extinguish a graphite fire, leaving leadership scrambling to adapt military assets for untested containment strategies.

Possible Solutions:

Delegates must now consider how best to physically contain the active disaster. Smothering the graphite fire will require the mobilization of the Soviet Air Force to conduct

high-risk helicopter drops of sand, clay, lead, and boron carbide directly into the exposed crater. To prevent the secondary thermal explosion, delegates must authorize a manual operation to drain the bubbler pools. This will require sending a specialized team of engineers into the flooded, highly radioactive basement levels to manually open the sluice gates—a mission with severe health implications for the volunteers.

Furthermore, the committee must establish a clear logistical framework for the triage of acute radiation victims and the immediate evacuation of the surrounding civilian population. By coordinating military, scientific, and medical resources efficiently, the committee can prevent the thermal explosion and set the stage for long-term containment.

Guiding Questions

1. What logistical measures must be taken to mobilize the materials (boron, lead, sand) and military aircraft necessary to smother the open reactor fire?
2. How will the committee authorize and execute the high-risk manual draining of the bubbler pools, and who will be selected for this operation?
3. What is the precise timeline and perimeter for the evacuation of Pripjat, and how will this be communicated to the public without inducing mass panic?
4. How should medical resources and advanced treatments be allocated to first responders currently suffering from Acute Radiation Syndrome?
5. How can the committee secure accurate radiation readings and diagnostic information from the core when standard equipment has been destroyed or irradiated?
6. In the event that helicopter containment drops fail, what secondary operational protocols can be deployed to suppress the reactor fire?

Topic B: The Groundwater Threat

Background:

If the immediate threats of the graphite fire and the bubbler pools are neutralized, the Chernobyl disaster enters a new, insidious, and largely invisible phase. The smothered reactor core, combined with the thousands of tons of sand, clay, and lead dropped by helicopters, creates a terrifying new problem. This mixture has morphed into "corium", a man-made, highly radioactive lava composed of melting uranium fuel rods, graphite, concrete, and containment materials.

This corium mass generates immense decay heat, sustaining temperatures upwards of 2,000 degrees Celsius. While the atmospheric spread of radiation has been reduced by smothering the fire, the danger has shifted underground. The corium is slowly burning through

the reactor's lower biological shield and the concrete foundation of the facility at a rate of several millimeters per hour. This phenomenon, often referred to as the "China Syndrome," presents a catastrophic ecological threat to the entire continent.

Directly beneath the Chernobyl Nuclear Power Plant lies the local water table, an aquifer that feeds directly into the Pripyat River. The Pripyat River flows into the Dnieper River, which serves as the primary water source for the capital city of Kyiv and the broader Ukrainian Soviet Socialist Republic. If the corium breaches the concrete foundation and contacts the aquifer, the resulting radioactive contamination will irreversibly poison the drinking water for over 30 million Soviet citizens and devastate the agricultural heartland of the USSR.

Current Situation:

The committee faces a rapidly closing window of opportunity to halt the subterranean descent of the corium. Surface-level cooling efforts have proven completely ineffective at lowering the temperature of the core mass. The sheer weight of the containment materials dropped on the reactor is exacerbating the structural stress on the compromised foundation. Radiation levels within the plant remain lethal, meaning any operation to address the melting core must be conducted in incredibly hazardous conditions. The Soviet scientific community is fiercely debating the exact timeline of the potential breach, but consensus dictates that action must be taken immediately before the lava reaches the earth below.

The state is forced to look toward extreme industrial solutions, requiring the rapid diversion of resources from the national economy. The environment is chaotic; leadership must balance the safety of the workers executing the containment strategy against the existential threat posed to the nation's water supply.

Past Actions:

The Soviet Union has a robust and highly disciplined industrial mining sector, which has historically been utilized for massive state-sponsored infrastructure projects. However, mobilizing this workforce for a subterranean nuclear containment operation is unprecedented. Previous actions taken during the initial days of the disaster focused entirely on fighting the fire from above, which inadvertently created the heavy, insulating cap that is now driving the corium downward.

There is a historical reliance on throwing sheer manpower at industrial problems within the USSR. In the context of Chernobyl, early attempts to assess the structural integrity of the lower levels involved sending unprotected soldiers into highly contaminated zones, resulting in severe casualties. The lack of remote-operated drilling or monitoring technology capable of surviving the intense gamma radiation has forced the state to rely entirely on human labor and analog solutions.

Possible Solutions:

To stop the corium from reaching the water table, delegates must engineer and authorize a massive subterranean mining operation. The only viable solution is to cool the foundation from below. The committee must mobilize a large workforce of skilled coal miners (likely conscripted from regions like Tula or the Donbas) to excavate a relief tunnel beneath Reactor No. 4.

Once the chamber is excavated, delegates must orchestrate the installation of a massive "heat exchanger" pad. This will require the procurement and continuous pumping of massive quantities of liquid nitrogen to artificially freeze the earth, followed by the pouring of heavily reinforced concrete to permanently seal the breach. The committee must carefully manage the labor force, addressing the lethal working conditions, high temperatures, lack of oxygen, and ambient radiation, while ensuring the operation is completed before the foundation gives way.

Guiding Questions

1. How will the committee quickly source, transport, and house the thousands of skilled miners required for the subterranean excavation?
2. What safety standards and radiation exposure limits, if any, will be enforced for the miners working directly beneath the melting core?
3. How will the massive quantities of liquid nitrogen and specialized concrete required for the heat exchanger be procured and transported to the Exclusion Zone?
4. What structural contingencies must be put in place in case the excavation process accelerates the collapse of the reactor foundation?
5. How will the state compensate or provide long-term healthcare for the mining workforce exposed to these extreme conditions?
6. If the subterranean cooling pad fails to halt the corium, what is the Soviet Union's backup plan to protect the Dnieper River basin from total contamination?

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Dossier:

1. **Mikhail Gorbachev** – General Secretary of the CPSU; served as a leading decision-maker within the party, guiding committee discussions and promoting *glasnost* reforms while working to preserve overall party stability.
2. **Nikolai Ryzhkov** – Soviet Premier; coordinated national emergency management and allocated state funds for the massive cleanup effort.
3. **Yegor Ligachev** – Senior Politburo member; opposed transparency and enforced strict party discipline to suppress alarmist reports.
4. **Viktor Chebrikov** – KGB Chairman; controlled information flow, monitored public reaction, and severed communications to prevent leaks.
5. **Eduard Shevardnadze** – Soviet Foreign Minister; managed international diplomatic fallout and negotiated foreign medical aid.
6. **Volodymyr Shcherbytsky** – First Secretary of Soviet Ukraine; governed the disaster zone and enforced public normalcy despite radiation risks.
7. **Boris Shcherbina** – Deputy Chairman; led the government commission on the ground and authorized immediate containment measures.
8. **Valery Legasov** – Lead scientist; investigated the reactor flaws and advocated for technical truth against political pressure.

9. **Anatoly Aleksandrov** – President of the Soviet Academy of Sciences; designed the RBMK reactor and defended its safety record to protect his legacy.
10. **Dr. Leonid Ilyin** – Chief radiologist of the USSR; established public health protocols and determined acceptable radiation exposure limits.
11. **Vasily Nesterenko** – Belarusian nuclear physicist; operated independent monitoring networks and demanded protection for northern regions.
12. **Vladimir Pikalov** – Commander of Chemical Troops; led initial reconnaissance missions and secured accurate radiation readings from the core.
13. **Nikolai Tarakanov** – Commander of Liquidators; mobilized reservists for debris removal and managed high-risk roof clearing operations.
14. **Yuri Samoilenko** – Deputy Chief Engineer; procured remote-controlled machinery and devised engineering solutions for decontamination.
15. **Alexei Ananenko** – Mechanical Engineer; volunteered to drain the bubbler pools beneath the reactor to prevent a thermal explosion.
16. **Viktor Bryukhanov** – Plant Director; responsible for overall operations and delayed reporting the full extent of the accident.
17. **Nikolai Fomin** – Chief Engineer; approved the fatal safety test and attempted to blame shift operators for the explosion.
18. **Anatoly Dyatlov** – Deputy Chief Engineer; supervised the failed safety test and denied the possibility of core destruction.
19. **Alla Yaroshinskaya** – Investigative journalist; uncovered government cover-ups regarding civilian radiation exposure and secret protocols.
20. **Yelena Bonner** – Human-rights activist; leveraged Western media connections to highlight the systemic failures of the Soviet regime.
21. **Olha Ivanivna** – Pripyat City Council representative; advocated for the immediate evacuation and medical needs of displaced residents.
22. **Gennadi Gerasimov** – Foreign Ministry spokesman; managed the global media narrative and deflected international inquiries.
23. **Dr. Robert Gale** – American hematologist; coordinated Western medical relief and treated acute radiation victims in Moscow.
24. **Hans Blix** – Director General of the IAEA; demanded international inspections and influenced global nuclear safety standards.
25. **Margaret Thatcher** – UK Prime Minister; utilized intelligence on the disaster to pressure the USSR in Cold War negotiations.